KASBench Full Disclosure Report

Preamble

This report documents the results obtained from an execution of the KASBench Kubernetes Autoscaling Benchmark.

Talk about completely random design.

Talk about run duration and number of trials

Insert a section for each autoscaler (how configured)

Benchmark Metadata

Benchmark Name	KASBench
Version	v1.0
Testing Dates	2026-09-18 20:33:12 UTC to 2026-09-19 10:16:10 UTC
Reporting Date	2026-09-24 23:42:36 UTC
Benchmark Application	GlobeCo
Cloud Provider	Amazon Web Services
Region	us-east-1
Benchmark Runner	c6a.large (amd64)
Control Plane	m8i.large (amd64)
Worker Nodes AMD	1 x c6a.4xlarge
Worker Nodes ARM	1 x c6g.4xlarge
AMD Worker AMI ID	ami-0244585558b2aebd7
ARM Worker AMI ID	ami-052bbac83b5bb0ab9
AMD Runner AMI ID	ami-03a891c9de365d954
Kubernetes Version	1.36.1
Autoscalers Evaluated	Kubernetes Horizontal Pod Autoscaler (HPA), Kubernetes Event-Driven Autoscaler (KEDA), Kubernetes Vertical Pod Autoscaler (VPA), No autoscaler
Trials Per Autoscaler	4

Software artifacts

Benchmark Components

Component	Image_original	ImageID
KASBench Controller	kasbench/kasbench-controller:0.0.6	sha256:0816294976161de3279756584c5ee4061eafb860d80e436ca0a33dc99cedabc4
KASBench Runner	kasbench/kasbench-runner:0.2.22	sha256:a28c2f8b2a9ca961e2186d0b5e078e373065427b406bc13752a98f9b84d63085
KASBench Load Generator	kasbench/kasbench-load-generator:0.0.21	sha256:5e6ad9d312140a7f6e19f5792aa68b4a1c12d1e2c6999e98f0bd5f24549bca4d

Helm Charts

Name	Version	URL
aws-ebs-csi-driver	2.66.0	https://kubernetes-sigs.github.io/aws-ebs-csi-driver
aws-efs-csi-driver	4.5.0	https://kubernetes-sigs.github.io/aws-efs-csi-driver/
globeco	0.1.51	https://kasbench.github.io/globeco-helm
keda	2.20.2	https://kedacore.github.io/charts
prometheus	29.31.1	https://prometheus-community.github.io/helm-charts

Stateful Services

Stateful Set	Image
postgres	postgres:18.0 docker.io/library/postgres sha256:41fc5342eefba6cc2ccda736aaf034bbbb7c3df0fdb81516eba1ba33f360162c
apache/kafka	apache/kafka:4.0.0 docker.io/apache/kafka sha256:3f7b939115cd4872e9cee9369d80bd69712fde55f9902f46d793f64848dedc75
mongo	mongo:6.0 docker.io/library/mongo sha256:8b6d8f5bbedb25cb73517b65cf99f13aeb75ad5b157a56c479287a840bbad3ac
redis	redis:7-alpine docker.io/library/redis sha256:520775a41a63e77e06c73e35d2fd9cc15921a609516818796b4ecbb813078bc7

Stateless Services

Deployment	Image
globeco-allocation-service	kasbench/globeco-allocation-service-server:1.0.3 docker.io/kasbench/globeco-allocation-service-server sha256:0e6750db5d9e6227554a346f12d0556ea4c62a6047786f2395a539c74218cd5e
globeco-confirmation-service	kasbench/globeco-confirmation-service:1.0.6 docker.io/kasbench/globeco-confirmation-service sha256:5d3267895a231b9210451f253b0ed783301520f036c4eab085b736405e3d7d64
globeco-debug-tools	kasbench/globeco-debug-tools:latest docker.io/kasbench/globeco-debug-tools sha256:9f279ca1ccaf4d118c3de3587bb7946198ac6a14569e3990ea978fdc4f644032
globeco-execution-service	kasbench/globeco-execution-service:1.0.7 docker.io/kasbench/globeco-execution-service sha256:07eb1ec41a44e1b6a80a50b17c4121372bd95a6c51879b08afbe79a28533c9e6
globeco-fix-engine	kasbench/globeco-fix-engine:1.0.13 docker.io/kasbench/globeco-fix-engine sha256:1705dd116250a607d73e1fab8bb6f44f67ea289fa848d4a841d64fe924c1e5d7
globeco-order-generation-service	kasbench/globeco-order-generation-service:1.0.6 docker.io/kasbench/globeco-order-generation-service sha256:a7541d05c1e720573394cb41b443067a05bbb7db75e69e240df1808c5fed5f14
globeco-order-service	kasbench/globeco-order-service:1.0.4 docker.io/kasbench/globeco-order-service sha256:1600b3c42ace32ddf0296468173bc835344e4c885e0b528f4d4e71e51f457b7f
globeco-portfolio-accounting-service	kasbench/globeco-portfolio-accounting-service-server:1.0.2 docker.io/kasbench/globeco-portfolio-accounting-service-server sha256:2b5646e0ccd9a5c4bf4bbce2749f6325c668802175cfc551c36c24fa36dee69d
globeco-portfolio-management-portal	kasbench/globeco-portfolio-management-portal:1.0.5 docker.io/kasbench/globeco-portfolio-management-portal sha256:a65a6022cb244c4bafoeac2ad1f8dddf626ec4ecf0481947e639a0924feb1193
globeco-portfolio-service	kasbench/globeco-portfolio-service:1.0.4 docker.io/kasbench/globeco-portfolio-service sha256:b0561599453606420eb4e904f31214c9c62c5a580c5ca95b25e59f67eca16dd7
	kasbench/globeco-portfolio-service-high-cpu:1.0.4 docker.io/kasbench/globeco-portfolio-service-high-cpu sha256:1a7821e35baf1814f2a7fddbd446ca59b2deaa86fe0a822a5400a27fb59be49c
globeco-pricing-service	kasbench/globeco-pricing-service:1.0.2 docker.io/kasbench/globeco-pricing-service sha256:2509c2a2e69712fa9a559b21a6f820ee537300592e6b77794c7135f282c06dc9
globeco-security-service	kasbench/globeco-security-service:1.0.1 docker.io/kasbench/globeco-security-service sha256:7066be0ea6f76bd60d1ba3c24a30e748c9c767a5eb13a2859204ab765e632241
globeco-trade-service	kasbench/globeco-trade-service:1.0.4 docker.io/kasbench/globeco-trade-service sha256:9c8f610f5fc34dac3eecf396a0f38f9ce4e24c22dea2b1a7acb7726962b26cb4

Raw Performance Metrics

Summarized Performance Metrics by Trial

Trial ID	Autoscaler	Requests	Mean Response Time (ms)	Failures	Failure Rate	Round-trip Completion (%)
trial0001	HPA	271092	100	21	0.008%	1.4%
trial0002	HPA	269018	96	10	0.004%	1.1%
trial0003	VPA	270496	79	11	0.004%	2.2%
trial0004	KEDA	263695	73	168	0.064%	6.2%
trial0005	NONE	259293	205	55	0.021%	1.7%
trial0006	VPA	268839	82	1929	0.718%	2.5%
trial0007	NONE	263339	165	195	0.074%	1.0%
trial0008	HPA	269791	96	21	0.008%	1.1%
trial0009	KEDA	271437	79	4	0.001%	5.9%
trial0010	VPA	269663	75	5	0.002%	2.5%
trial0011	VPA	264706	77	734	0.277%	2.3%
trial0012	NONE	192960	188	16	0.008%	1.8%
trial0013	KEDA	263246	102	130	0.049%	6.8%
trial0014	HPA	270773	84	8	0.003%	1.2%
trial0015	KEDA	270186	94	112	0.041%	6.2%
trial0016	NONE	214979	240	165	0.077%	2.5%

Summarized Performance Metrics by Autoscaler

Autoscaler	Mean Response Time (ms)	Failure Rate	Round-trip Completion (%)
HPA	94	0.006%	1.2%
KEDA	87	0.039%	6.3%
NONE	200	0.045%	1.7%
VPA	78	0.250%	2.4%

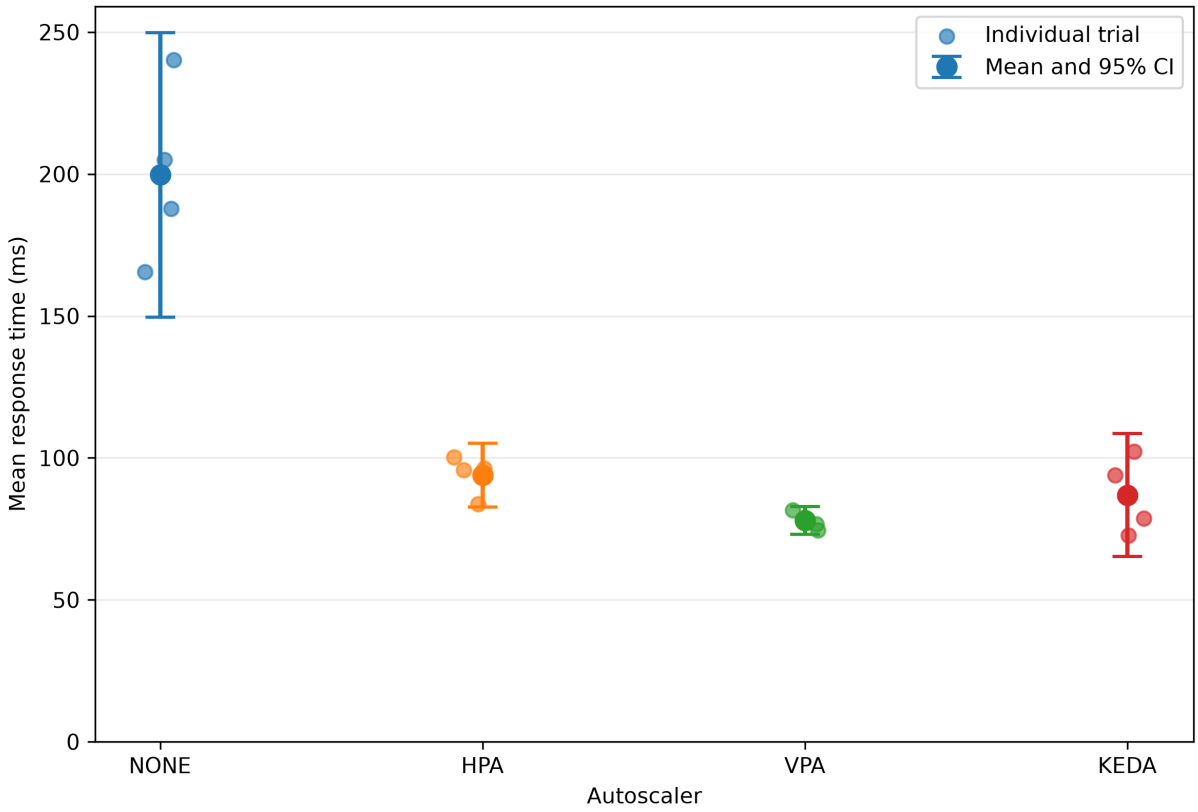


Figure 1. Mean response time by autoscaler.

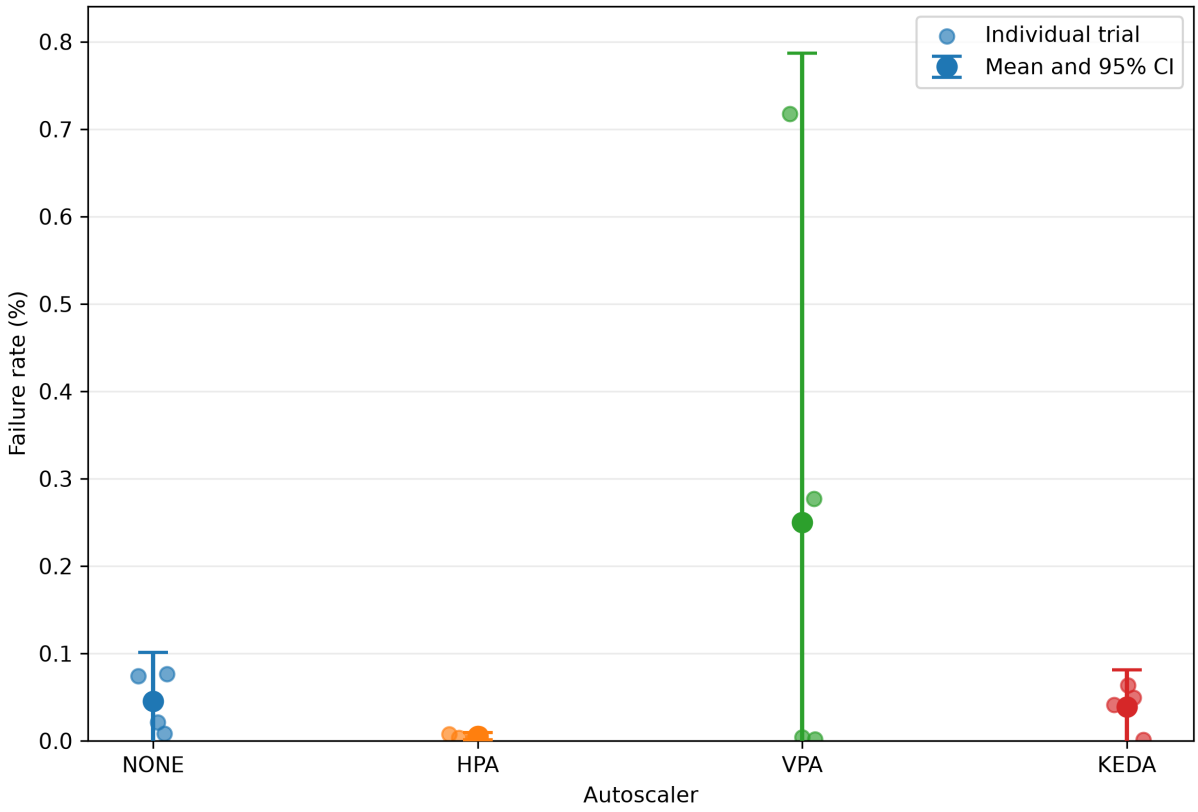


Figure 2. Mean failure rate by autoscaler.

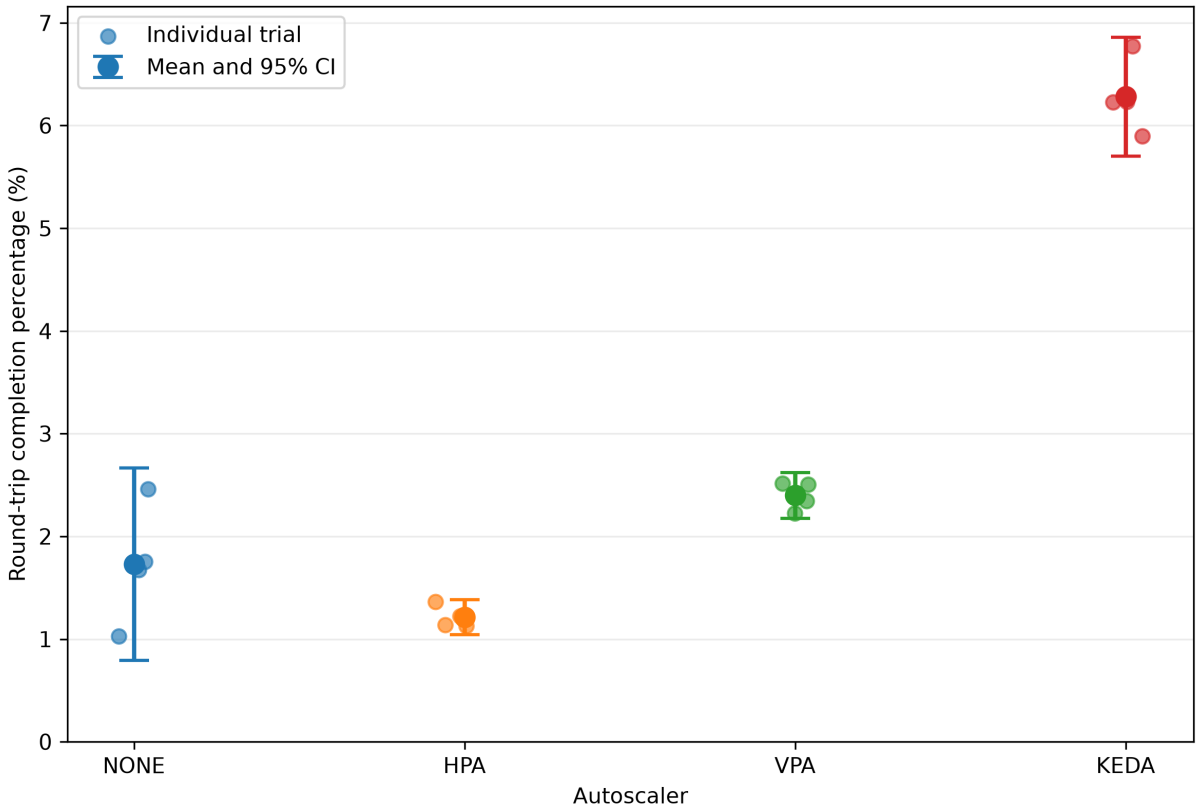


Figure 3. Mean round trip percentage by autoscaler.

Comparative Autoscaler Metrics

Synchronous Misprovisioning Penalty (θ^{S}) by Autoscaler

Autoscaler	θ_U^{RT}	θ_O^{RT}	θ_U^{FR}	θ_O^{FR}	θ^{RT}	θ^{FR}	θ^{S}
HPA	240.030	1.213	250.149	91.588	120.621	170.869	145.745
KEDA	211.628	1.710	2247.561	88.541	106.669	1168.051	637.360
NONE	635.153	0.722	1268.497	73.100	317.938	670.799	494.368
VPA	173.943	1.261	1426.147	92.412	87.602	759.280	423.441

Synchronous Misprovisioned Timeshare (τ^S) by Autoscaler

Autoscaler	τ_U^{RT}	τ_O^{RT}	τ_U^{FR}	τ_O^{FR}	τ^{RT}	τ^{FR}	τ^S
HPA	88.934	0.820	4.918	92.213	44.877	48.566	46.721
KEDA	86.885	3.689	8.197	89.754	45.287	48.975	47.131
NONE	92.213	1.230	18.443	75.410	46.721	46.926	46.824
VPA	88.115	2.869	4.098	94.262	45.492	49.180	47.336

Asynchronous Metrics (θ^A and τ^A) by Autoscaler

Autoscaler	Avg. Lag	θ_U^A	θ_O^A	τ_U^A	τ_O^A	θ^A	τ^A
HPA	6874.297	34319.223	41.911	49.593	6.794	17180.567	28.194
KEDA	50.250	208.170	42.187	23.843	19.298	125.178	21.570
NONE	3853.360	19217.293	42.618	46.382	13.846	9629.956	30.114
VPA	183.786	875.060	52.729	36.226	7.809	463.894	22.017

Cost Metrics by Autoscaler

Autoscaler	$\overline{C}_{CPU}$	$\overline{C}_{MEM}$	$\overline{C}$
HPA	86.690	69.573	78.131
KEDA	92.114	73.469	82.791
NONE	55.694	45.026	50.360
VPA	79.545	63.998	71.771

Summary Results by Autoscaler

Autoscaler	$\overline{\theta}^S$	$\overline{\tau}^S$	θ^A	τ^A	ζ	$\overline{C}$
HPA	145.745	46.721	17180.567	28.194	4295.142	78.131
KEDA	637.360	47.131	125.178	21.570	159.400	82.791
NONE	494.368	46.824	9629.956	30.114	2407.493	50.360
VPA	423.441	47.336	463.894	22.017	132.315	71.771